DESIGN AND CONCEPTUAL ARCHITECTURE OF AN IN-VEHICLE INFOTAINMENT SYSTEM

Integrating Media, Navigation and Smartphone Connectivity

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Abstract

An In-Vehicle Infotainment (IVI) system acts as a central software platform inside a modern vehicle, bringing entertainment, navigation, communication and connected services together through a common interface. This project develops a conceptual architecture for an IVI platform containing three core areas: media playback, navigation and smartphone projection. The proposed design divides the system into user interaction, application services, middleware, hardware abstraction and external interfaces. This separation explains how a command from the driver reaches the appropriate service and is eventually converted into audio, visual or connectivity output. The report also examines shared-resource management, particularly audio focus, so that navigation announcements or phone calls can temporarily interrupt media without creating conflicts. The model is intended as an educational system-level design and can serve as a foundation for a future implementation using Android Automotive or a comparable embedded platform.

1. Introduction

Modern vehicles increasingly depend on software to provide entertainment, navigation, communication and access to online information. An IVI system provides a unified environment through which passengers and drivers interact with these functions. Unlike a standalone application, an IVI platform must coordinate several features while they share resources such as the display, speakers, microphone, location receiver and wireless communication interfaces.

A layered architecture makes this coordination easier to understand. Application features remain separated from common system services and hardware-specific components. As a result, the same application logic can work with different hardware implementations, while shared services can control resources that are used by multiple features.

2. Aim and Objectives

The main aim of this project is to develop a conceptual system architecture for an IVI platform capable of handling media, navigation and smartphone projection.

The objectives are:
• Identify the major functional blocks required in an IVI system.
• Organize the system into clear software and hardware-related layers.
• Describe the communication paths between applications, middleware and hardware.
• Explain the handling of audio, location, vehicle and smartphone data.
• Demonstrate how shared resources are coordinated when multiple events occur.
• Provide a foundation that could later be converted into a working prototype.

3. Scope of the Project

The work focuses on system-level design rather than implementation of production software. It covers the human-machine interface, media and navigation services, smartphone projection, common middleware, hardware abstraction interfaces and external components such as speakers, GNSS, CAN-based vehicle signals, smartphones and network services.

Safety-critical automated-driving functions are not included. Similarly, production-grade security, authentication, diagnostics and OEM-specific features are considered outside the present scope.

4. Functional Requirements

The proposed system should provide the following functions:

- Accept commands through the touchscreen, steering-wheel controls and voice input.

• Play music and manage playlists, metadata, playback state and volume.

• Provide maps, route calculation, traffic information, current location and spoken directions.

• Establish smartphone connectivity through USB, Wi-Fi or Bluetooth.

• Coordinate audio priority so that navigation instructions and calls can temporarily reduce or interrupt media playback.

• Access vehicle-related information through a Vehicle HAL/CAN interface.

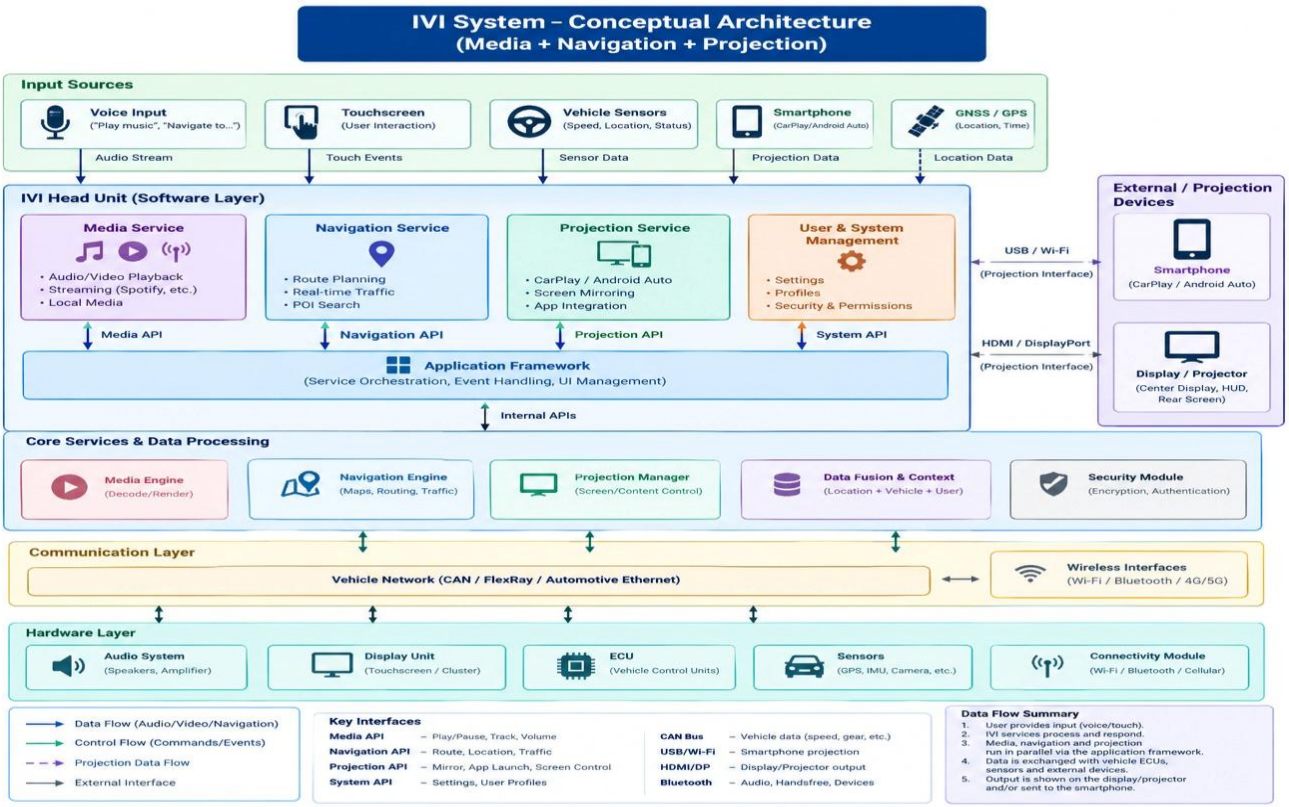
• Render application information on the vehicle display and send final audio streams to the amplifier and speakers.

5. Proposed System Architecture

The IVI platform is organized as a stack of cooperating layers. User-facing interactions are handled at the top, while hardware and external services are located at the lower levels. Middleware acts as the coordination layer between applications and the underlying platform.

The major layers are:
1. User Interaction and HMI
2. Application / Feature Services
3. Middleware and System Services
4. Hardware Abstraction and Platform Interfaces
5. External Hardware and Online Services

This arrangement reduces direct dependency between applications and hardware and makes the overall system easier to extend.



6. Major Components

6.1 User Interaction and HMI

The Human-Machine Interface is the part visible to the vehicle occupants. It includes the touchscreen, display, steering-wheel controls, microphone and voice interaction. User actions are converted into software events, while application results are displayed or sent to the audio system.

6.2 Application and Feature Layer

The Media Service controls playback, playlists, metadata and related user actions. The Navigation Service manages route guidance, map information, traffic data and spoken instructions. The Projection Manager handles a connected smartphone and manages supported projected applications. A telephony event can also request priority access to the audio system.

6.3 Middleware and System Services

Middleware provides reusable services required by multiple applications. The Audio Manager controls volume, routing, mixing and audio focus. Activity or Window Management handles application lifecycle and display windows. Notification services present calls and alerts. Bluetooth and Connectivity Managers maintain wireless and network connections. Location services provide GNSS information, while the media framework manages media sessions, codecs and metadata.

6.4 Hardware Abstraction Layer

The Hardware Abstraction Layer separates software components from hardware-specific implementations. The Audio HAL converts software audio streams into hardware output. Vehicle HAL exposes vehicle information obtained through CAN or other vehicle interfaces. GNSS interfaces provide positioning data, while display, touch, USB, Wi-Fi and Bluetooth drivers connect software functions with physical devices.

6.5 External Components

The external environment includes the vehicle speakers and amplifier, sensors and CAN network, GNSS satellites, smartphones and cloud/network services. These components provide data to the IVI system or receive processed information from it.

7. Interfaces and Data Exchange

Different services communicate through well-defined interfaces. The HMI sends touch, steering and voice events to application services. Media playback information is passed to the Audio Manager and Media Framework, which coordinate the session and audio policy. The Audio Manager forwards the resulting stream to the Audio HAL, which sends it to the amplifier and speakers.

Navigation obtains position information from the Location Manager and can request temporary audio focus when a voice instruction is required. The Projection Manager communicates with the Projection Service to establish a smartphone session, while USB, Wi-Fi or Bluetooth carry projection data. Vehicle information reaches the software stack through the Vehicle HAL and CAN/sensor interfaces. Connectivity services provide online maps, traffic information and other network-based content.

8. End-to-End Operation

8.1 Media Playback

When the user selects a song, the HMI generates an input event and passes it to the media service. The service updates the media session and requests audio output. The Audio Manager applies the required volume, routing

and focus rules. The stream then passes through the Audio HAL to the amplifier and speakers. Playback information such as the current track can simultaneously be returned to the display.

8.2 Navigation

The user enters or selects a destination. The navigation service obtains current location and, where required, online map or traffic information. After calculating the route, it presents guidance on the HMI. For a spoken instruction, the navigation service requests temporary audio focus. Existing media audio is then reduced or paused, the instruction is played, and normal media playback can resume.

8.3 Smartphone Projection

A smartphone is connected using an available interface such as USB, Wi-Fi or Bluetooth. The projection service establishes the communication path and the projection manager creates a session. Supported phone applications and their data are presented through the vehicle display. Touch or steering-wheel commands can be transferred back to the smartphone, while compatible media, call and navigation audio can use the vehicle's audio system.

9. Cross-Feature Scenario: Incoming Phone Call

Consider a situation in which music is playing while navigation is active. When an incoming call is detected, the telephony and notification components request higher-priority audio access. The Audio Manager temporarily suppresses or pauses media and routes the call audio to the vehicle speakers. Navigation information can remain visible, while spoken guidance can be handled according to the system's audio policy.

After the call ends, the call-related component releases audio focus. The media service receives the updated focus state and can continue playback. This example shows why an IVI system needs shared middleware: media, navigation and telephony cannot independently control the same audio resources without coordination.

10. Advantages of the Architecture

- The layered structure separates user applications from hardware-dependent implementation details.
- Common middleware allows several features to share audio, display, connectivity and location resources.
- Media, navigation and projection can be developed as separate functional modules.
- Hardware abstraction makes the design adaptable to different vehicle platforms.
- The architecture provides a clear method for handling temporary interruptions such as calls and navigation prompts.
- The design can be expanded toward a more complete connected-cockpit platform.

11. Limitations and Future Work

The proposed design is conceptual and does not represent a complete production automotive system. Areas such as functional safety, cybersecurity, user authentication, over-the-air software updates, diagnostics, power management, multi-display synchronization and manufacturer-specific services are not implemented.

As future work, the architecture could be prototyped using Android Automotive OS or a similar platform. Mock vehicle properties could be connected through Vehicle HAL, while an emulator or development board could be used to observe application interactions and system logs. Additional work could also include security controls, diagnostic services and more detailed vehicle-data integration.

12. Conclusion

This project presents a conceptual architecture for an In-Vehicle Infotainment system that combines media playback, navigation and smartphone projection within a common platform. The layered approach shows how user commands travel from the HMI through application services, shared middleware and hardware abstraction before reaching physical devices or external services.

The design also demonstrates the importance of resource coordination through examples such as navigation prompts and incoming calls. Overall, the proposed architecture provides a structured basis for understanding connected-cockpit software and can be extended into a practical prototype on an automotive software platform.

13. References

1. Android Developers — Android for Cars and Android Automotive OS documentation.
2. Android Open Source Project (AOSP) — Automotive and Audio Architecture documentation.
3. Android Open Source Project (AOSP) — Vehicle HAL and automotive framework documentation.
4. Bluetooth SIG — Bluetooth specifications and profiles for media and hands-free connectivity.
5. General concepts of automotive infotainment and connected-cockpit system architecture.